

Customer Shopping Behavior Analysis

1. Project Overview

An integrated analysis of 3,900 e-commerce records using Python, MySQL, and Power BI. This project focuses on the intersection of customer demographics and purchasing habits, specifically engineering features to isolate 'Loyal' vs 'New' shoppers. The final output is an interactive intelligence suite designed to improve customer retention and revenue forecasting.

2. Dataset Summary

This analysis is driven by a high-granularity dataset containing **3,900 unique consumer interactions**. The schema is composed of **18 strategic variables** which I have categorized into three primary analytical domains:

- User Demographics:** Includes partitioned age groups, gender distribution, geographic location, and subscription tiers.
- Transaction Mechanics:** Detailed logs of item specifications (category, size, colour) and seasonal timing.
- Behavioral Metrics:** Quantitative indicators of loyalty, such as purchase frequency, historical volume, and engagement with promotional discounts.
- Data Integrity:** During the initial audit, I identified 37 null entries within the 'Review Rating' field. These were successfully resolved via median imputation during the preprocessing phase in Python to ensure statistical accuracy in the final visuals.

3. Exploratory Data Analysis using Python

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900.000000	3900	3900
2	NaN	6	7
No	NaN	PayPal	Every 3 Months
2223	NaN	677	584
NaN	25.351538	NaN	NaN
NaN	14.447125	NaN	NaN
NaN	1.000000	NaN	NaN
NaN	13.000000	NaN	NaN
NaN	25.000000	NaN	NaN
NaN	38.000000	NaN	NaN
NaN	50.000000	NaN	NaN

We began with data preparation and cleaning in Python:

- **Data Acquisition:** Utilized the **Pandas** library to ingest the raw transactional dataset.
- **Structural Audit:** Executed `df.info()` and `.Describe()` to map the data architecture and establish baseline summary statistics.
- **Missing Value Resolution:** Identified null entries in the **Review Rating** column and applied a category-specific median imputation to maintain data integrity.
- **Naming Convention Standardization:** Transformed all column headers into **snake_case** to improve documentation clarity and coding efficiency.
- **Feature Engineering:**

Developed an **age_group** dimension by binning customer ages into logical segments.

Derived a **purchase_frequency_days** metric to better track customer re-engagement cycles.
- **Redundancy Audit:** Conducted a consistency check between `discount_applied` and `promo_code_used`; determined they were redundant and streamlined the dataset by dropping the latter.
- **Database Migration (ETL):** Established a secure connection to the database and loaded the refined DataFrame into **SQL** for advanced relational querying.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

The Finding: Male customers are the primary revenue drivers, contributing **\$157,890**—more than double the revenue of the female segment.

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
gender	revenue		
Male	157890		
Female	75191		

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

The Finding: Filtered the database for customers who used a discount code but still spent more than the average amount of \$59.76.

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
customer_id	purchase_amount		
2	64		
3	73		
4	90		
7	85		
9	97		
12	68		
13	72		
16	81		
20	90		
22	62		
24	88		
29	94		
32	79		
33	67		
35	91		

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

The Finding: Gloves and Sandals are the "Gold Standard" products, consistently maintaining satisfaction ratings above **3.8/5.0**.

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
item_purchased	review_rating			
Gloves	3.86			
Sandals	3.84			
Boots	3.82			
Hat	3.8			
Skirt	3.78			

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

The Finding: Customers choosing **Express Shipping** actually spend more per order (\$60.48 avg) than those using Standard methods.

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	shipping_type	ROUND(AVG(purchase_amount), 2)			
▶	Express	60.48			
	Standard	58.46			

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

The Finding: While non-subscribers drive the most volume, the **\$170,436** they spend is nearly identical in "per-order value" to subscribers, showing a massive opportunity to convert them without losing margin.

Result Grid					Filter Rows:	Export:	Wrap Cell Content:
	subscription_status	total_customers	average_spent	total_revenue			
▶	No	2847	59.87	170436			
	Yes	1053	59.49	62645			

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

The Finding: **Hats and Sneakers** are highly promotion-dependent, with nearly half of all sales tied to a discount.

Result Grid			Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	item_purchased	discount_rate				
▶	Hat	50.00				
	Sneakers	49.66				
	Coat	49.07				
	Sweater	48.17				
	Pants	47.37				

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

The Finding: The business is built on a rock-solid foundation of **3,116 "Loyal" customers**, providing highly predictable recurring revenue.

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
customer_segment	Number of Customers		
Loyal	3116		
Returning	701		
New	83		

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

The Finding: **Jewelry, Blouses, and Sandals** are the "anchor" products for the Accessories, Clothing, and Footwear categories.

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

The Finding: There are **2,518 frequent buyers** who are not yet subscribers—this is the #1 growth opportunity for the business.

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
subscription_status	repeat_buyers		
Yes	958		
No	2518		

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

The Finding: The **Young Adult** segment is the most profitable demographic, contributing a peak revenue of **\$62,143**.

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	age groups	total_revenue	
▶	young adults	62143	
	mid aged	59197	
	adults	55978	
	senior	55763	

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers. Targeted Subscription Conversion: Launch a specialized email campaign for the 2,518 repeat buyers identified in the SQL analysis who are not yet subscribers. Offering a "First-Month Free" trial to this high-frequency group is the #1 opportunity for growing recurring revenue.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the "Loyal" segment. Implement a structured rewards program to transition "New" and "Returning" shoppers into the "**Loyal**" segment. Focusing on the second and third purchase milestones will stabilize the long-term customer base
- **Review Discount Policy** – Balance sales boosts with margin control. Refine the discount policy for high-sensitivity categories like **Hats and Sneakers**. By shifting from flat discounts to "bundle-and-save" offers, the business can maintain sales volume while protecting profit margins.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns. Feature "Customer Favorites"—specifically high-rated items like **Gloves and Sandals**—in the primary hero sections of the website and marketing materials to leverage social proof.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users. Reallocate the digital advertising budget toward the **Young Adult** demographic and **Express Shipping** users. These segments have demonstrated the highest "Willingness to Pay," offering a superior Return on Ad Spend (ROAS).

